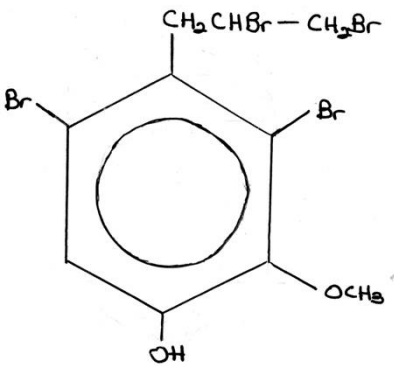


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
		(iii)		% bromine by mass = 66.4 (1) dividing by A_r (1) then dividing each by the smallest figure gives ratio leading to empirical formula of $C_5H_5Br_2O$ (1) compound contains two oxygen atoms / 10 carbon atoms so molecular formula is $C_{10}H_{10}Br_4O_2$ (1) possible structure is  (1)	1					
		(iv)		eugenol is acidic (phenol) and will react with NaOH to give a salt (which is soluble in the aqueous layer) (1) eugenyl ethanoate is an ester and is not soluble in the aqueous layer (1)			2			2
				Total question 10	3	5	7	3	2	15